

PROJECT BLUEPRINT

MULTIMODAL RAG-BASED AGRICULTURAL EXTENSION SYSTEM

Supporting Nigerian Pidgin and Local Dialects (Hausa, Igbo, Yoruba)

1. Project Overview & Problem Statement

Agricultural extension services serve as the backbone of food security in Nigeria. However, the extension agent-to-farmer ratio is extremely high (~1:5000), preventing farmers from obtaining real-time advice. This blueprint details the development of a Multimodal Retrieval-Augmented Generation (RAG) system specifically designed for Nigerian smallholder farmers, overcoming three major barriers:

- **Language Barrier:** Most digital tools are in English, whereas most smallholder farmers communicate in Nigerian Pidgin, Hausa, Igbo, or Yoruba.
- **Input Complexity:** Traditional text interfaces fail users with low literacy or those in field conditions where sending a voice note or taking a photo of a diseased leaf is more practical.
- **Real-Time Information:** Farmers lack immediate, expert-verified advice on pest control, disease treatment, and soil management, leading to crop losses.

Aim & Core Objectives:

To develop a Multimodal RAG-based information retrieval system providing localized, expert-verified agricultural support in Nigerian Pidgin and major local dialects.

- Design a multimodal vector database containing Nigerian-specific agricultural data (crop manuals, pest controls).
- Implement a RAG pipeline that fetches context-aware information based on text, image, and voice queries.
- Integrate translation modules (NLP) to handle the dialect-to-English translation loop step-by-step.
- Optimize the system for high performance, low latency, and offline capability on slower 3G networks.

2. System Architecture Blueprint

The architecture of the Agricultural RAG system comprises four core layers connected in an orchestrated pipeline. This modular approach separates translation, vector indexing, retrieval, and synthesis.

A. Multimodal Embedding Layer

- Image Processing (CLIP): Converts crop disease photos (e.g., infected Cassava leaves) into vector embeddings for similarity lookup against pre-diagnosed symptoms.
- Audio Processing (Whisper): Transcribes voice notes in Nigerian Pidgin or dialects and passes the transcripts to the translation module.

B. Vector Database Layer (ChromaDB)

ChromaDB stores high-dimensional vector representations of local agricultural manuals collated from FMAFS and IITA. Text queries and image embeddings are queried against the DB to retrieve the top-K relevant document snippets.

C. Translation Layer (NLP / NLLB)

A fine-tuned No Language Left Behind (NLLB) model manages the dialect-to-English translation loop. Since agricultural publications are primarily written in English, dialect queries are translated to English for vector matching and LLM prompting, and the resulting LLM response is translated back into the farmer's source dialect.

D. RAG Orchestration & LLM Generator

A middleware orchestrator combines the translated query with retrieved context from ChromaDB. It builds a structured prompt for the LLM to generate an accurate, hallucination-free agricultural recommendation.



3. Prototype Implementation Plan

The project will be built as an interactive dashboard prototype using React + TypeScript + Vite. It provides a visual simulator for all hardware-heavy components (NLLB translation, ChromaDB vector indexing, Whisper audio, and CLIP image parsing) running completely in-browser. This ensures high accessibility for presentations and testing.

Project Structure (Components & Services):

src/services/mockDatabase.ts

Pre-loaded agricultural knowledge base on local pests (Cassava Mosaic Virus, Maize Rust, Tomato Blight) and soil management guidelines.

src/services/ragPipeline.ts

Local engine simulating Whisper transcription, CLIP image analysis, ChromaDB similarity vector matches, and NLLB translation loops.

src/components/ChatAssistant.tsx

Main chatbot interface with a fluid multi-language picker, microphone recorder simulation, and image attachment options.

src/components/VectorDBExplorer.tsx

Vector database visual explorer allowing search querying, viewing similarity scores, and inspectable document chunks.

src/components/PipelineVisualizer.tsx

Interactive node graph demonstrating RAG stages. Shows backend metrics: latency, model confidence, and input/output language tokens.

src/components/LeafDiagnostic.tsx

Visual crop pathology diagnostic view with bounding-box overlay showing diseased spots on uploaded crop images.

Aesthetics & User Experience:

- Design Theme: Organic nature palette (forest green, mint green, warm earthy sand highlights, charcoal text) to convey growth and agricultural relevance.
- Layout: Mobile-first responsive grid. Interfaces are designed for low-literacy farmers (using rich icons, color-coded diagnostic alerts, and clear audio prompts).
- 3G Network Simulator: Integrates an adjustable network throttle to simulate high latency, illustrating how the app pre-fetches or buffers text/audio to support 3G field connections.

4. Future Integration & Production Roadmap

Transitioning from the interactive prototype to a fully integrated production system involves deploying dedicated cloud API endpoints for the heavy-duty model inferences.

PRODUCTION DEPLOYMENT COMPONENT BLUEPRINT		
Component	Prototype Mock	Production Architecture
Vector Database	In-Memory JS/JSON Search	ChromaDB Docker Server on AWS/GCP
NLLB Translation	Mapping dictionary / Mock NLLB	HuggingFace Inference API (nllb-200)
Audio Processing	Simulated visual waveform & text	OpenAI Whisper API / Speech-to-Text
Leaf Diagnostic	CLIP bounding-box image mockup	Roboflow YOLOv8 API / Custom CLIP Model
RAG Orchestrator	Client-side TS workflow runner	Python FastAPI Server / LangChain
LLM Generator	Context-anchored advisor mock	Google Gemini 1.5 Pro / GPT-4o API

RECOMMENDED TESTING METRICS FOR PHASE 1
• RAG Context Recall: Target >85% document relevance matching. • Translation BLEU Score: Verify translation quality for Nigerian Pidgin. • Total End-to-End Latency: Target < 4 seconds on 3G simulated network speeds. • Disease Diagnostics (F1 Score): Target >90% precision on leaf bounding box alerts.